



12116 - Line Radiative Transfer from Black Hole Envelopes: Detailed Models of JWST Little Red Dots

Cycle: 5, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The discovery of ubiquitous compact sources at high-redshift ("little red dots" or LRDs) is one of the most interesting early discoveries of JWST. Their red continuum color in the absence of dust, very broad emission lines and lack of X-rays are qualitatively explained as an envelope of material accreting quasi-spherically at a super-Eddington rate onto a growing central massive black hole. We propose to carry out the first radiative transfer calculations of the continuum and line spectra from such configurations for comparison with rich JWST spectra from a sample of LRDs. We will do this using non-LTE stellar atmospheres codes for fluid backgrounds corresponding to low- and high-mass ("quasi-star") envelopes, and test the models using key spectral diagnostics indicative of their physical properties which can be robustly measured from JWST data.

OBSERVING DESCRIPTION